An exact counterexample to trigonometric positivity for square roots of geometric sums

Research proof and formalization packet

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Abstract

We disprove Kovalev's Conjecture 1.3 on positive trigonometric partial sums of the Taylor coefficients of $\sqrt{1+z+\cdots+z^d}$. An exact counterexample has $d = 49$, partial-sum index 949, and $t = \arccos(523/527)$. A finite integer certificate proves the sum is strictly negative. The normalized square-root definition, coefficient recurrence, integer certificate, cosine identity and final source counterexample are all checked by Lean. The separate zero-free Taylor-polynomial conjecture is not refuted by this example.

1 The source conjecture and counterexample

For $d \geq 2$, define real coefficients $b_k^{(d)}$ by the branch at the origin with value one:

$$\sqrt{1+z+\cdots+z^d} = \sum_{k\geq 0} b_k^{(d)} z^k.$$

Kovalev's Conjecture 1.3 [1], indexed as MathDB 369172, states that

$$\mathcal{T}_n^{(d)}(t) = \sum_{k=0}^{n} b_k^{(d)} \cos(kt) > 0 \qquad (d \geq 2,\ n \geq 0,\ t \in \mathbb{R}).$$

The statement is present both in arXiv:2507.04544v1 and in the published 2026 paper, page 274.

Theorem 1. *With $t_0 = \arccos(523/527)$,*

$$\mathcal{T}_{949}^{(49)}(t_0) < 0.$$

Its approximate value is -0.000475321859370024; the proof below uses exact integers, not rounding. This is one counterexample outcome. The index, rational angle, auxiliary recurrences and certificate receive no separate discovery credits. Conjecture 1.2 in the same source concerns zero-free Taylor polynomials. Negating a sufficient positivity condition does not negate that different conjecture, and no such claim is made.

2 Identifying the source coefficients exactly

Write $D = d+1$, $G_D(z) = 1+z+\cdots+z^{D-1}$, and let F_D be its normalized formal square root. It exists, for example, by substituting $G_D - 1$ in the formal binomial series with exponent $1/2$. The identities $F_D(0) = 1$ and $F_D^2 = G_D$ also select it uniquely: two such roots have $(F_1-F_2)(F_1+F_2) = 0$

in the integral domain of formal series; the second factor has constant term two and cannot vanish. Thus these are exactly the Taylor coefficients specified in the source.

Since $F_D^2(1-z) = 1 - z^D$, formal differentiation and multiplication give

$$2(1-z)(1-z^D)F_D' = \big(1 + (D-1)z^D - Dz^{D-1}\big)F_D. \tag{1}$$

Writing $F_D = \sum b_k z^k$ and comparing the coefficient of z^{k-1} yields, for $k \geq 1$,

$$\begin{aligned} 2kb_k = (2k-1)b_{k-1} &+ \mathbf{1}_{k\geq D}(2k-3D)b_{k-D} \\ &- \mathbf{1}_{k\geq D+1}(2k-3D-1)b_{k-D-1}. \end{aligned} \tag{2}$$

Terms with negative indices are omitted. Since $2k \neq 0$, the initial value $b_0 = 1$ and this recurrence determine every needed coefficient.

For $D = 50$, the packet supplies integer data $B_0, \ldots, B_{949}$ and checks $B_0 = 1$ and, for $1 \leq k \leq 949$,

$$\begin{aligned} 2kB_k = 4(2k-1)B_{k-1} &+ \mathbf{1}_{k\geq 50}4^{50}(2k-150)B_{k-50} \\ &- \mathbf{1}_{k\geq 51}4^{51}(2k-151)B_{k-51}. \end{aligned} \tag{3}$$

By induction using (2), these finite integer checks prove $B_k = 4^k b_k^{(49)}$ throughout the required range. No guessed coefficient values or unproved recurrence assumptions enter the final source theorem.

3 A finite integer sign certificate

Let $p = 523$, $q = 527$, and define integers

$$C_0 = 1, \quad C_1 = 523, \quad C_k = 1046C_{k-1} - 277729C_{k-2} \quad (k \geq 2).$$

The cosine addition identity and $\cos t_0 = p/q$ give, by induction,

$$C_k = q^k \cos(kt_0).$$

Our finite C table is checked against these exact initial values and recurrences through index 949.

Finally put

$$H_0 = 0, \qquad H_{k+1} = 2108H_k + B_kC_k \quad (0 \leq k \leq 949).$$

Induction in Horner's recurrence gives

$$H_{950} = 2108^{949} \sum_{k=0}^{949} \frac{B_k}{4^k}\frac{C_k}{527^k} = 2108^{949}\mathcal{T}_{949}^{(49)}(t_0). \tag{4}$$

The certificate computes $H_{950} < 0$ in exact integer arithmetic. Its absolute value has 3152 decimal digits; the complete integer and denominator 2108^{949} are included in the packet rather than printed as several pages of digits. Since the multiplier in (4) is positive, this proves Theorem 1.

For reproducibility, the finite data can be generated from

$$(1-z^{50})^{1/2}(1-z)^{-1/2} = \left(\sum_{j\geq 0} a_j z^{50j}\right)\left(\sum_{r\geq 0}\frac{\binom{2r}{r}}{4^r}z^r\right),$$

where $a_0 = 1$ and $a_j = a_{j-1}(2j-3)/(2j)$. The generator uses only rational arithmetic. Its correctness is not a trusted premise: Lean independently checks (3), the C recurrences and the negative Horner result for the literal integer data.

4 Formal verification and prior-art review

The seven-module Lean certificate has 55 inventoried declarations. The source root is constructed unconditionally from Mathlib's formal binomial series. The source-to-recurrence bridge and finite recurrence uniqueness identify its coefficients with the certified data. The final endpoint `KovalevCounterexample.explicit_source_counterexample` has no assumptions. The companion endpoint formally negates the universal positivity conjecture. No analytic remainder bound, floating-point sign decision or missing source hypothesis is assumed.

The integer checks use ordinary kernel-checked `decide`; no `native_decide`, added axiom or proof hole is used. The complete certificate replays in pinned Lean 4.34.1. The only reported axioms are `propext`, `Classical.choice`, and `Quot.sound`. The tables are encoded by balanced integer lookups to keep replay practical; this affects representation, not the statement or trust model.

Independent exact computations by direct square-root convolution and scaled integer or quadratic-ring cosine arithmetic agree with the negative value. A separate reviewer checked the mathematical and source bridges, including $d = 49$ versus $D = 50$ and the H_{950} indexing. The actual public publisher PDF was inspected and retains Conjecture 1.3. Documented literature searches located no prior counterexample to this exact statement. This is a finite novelty investigation, not an absolute certification of worldwide priority. No minimality in d or n is claimed.

References

[1] L. V. Kovalev, *Sharp bounds for some segments of bounded power series*, Arkiv för Matematik 64(2) (2026), 273–282, Conjecture 1.3. DOI: 10.4310/ARKIV.2026.v64.n2.a4. Preprint: arXiv:2507.04544v1 (2025).